VIBRATION—FELT THROUGH STEERING WHEEL AND/OR SEAT AT SPEEDS ABOVE 72 KM/H (45 MPH)	Article No. 01-22-2

MERCURY: 1999-2000 VILLAGER

This article is being republished in its entirety to update the Service Procedure.

ISSUE

A slight vibration/shake may be felt on some vehicles in the steering wheel and/or floor pan and front seat at steady vehicle speeds above 72 km/h (45 mph) on smooth roads after the tire/wheel assemblies are balanced. This condition may be caused by the ride characteristics of one or more of the Original Equipment Manufacturer (OEM) tires.

ACTION

To resolve a vibration/shake felt in the steering wheel and/or floor pan and front seat at vehicle speeds above 72 km/h (45 mph) that was not corrected with wheel/tire balancing, isolate the tire(s) and replace. Refer to the following Evaluation and Service Procedure for details.

EVALUATION

NOTE
THIS PROCEDURE APPLIES ONLY TO VILLAGER SPORT AND ESTATE MODELS EQUIPPED WITH P225/60R16 GOODYEAR EAGLE LS TIRES.

Driveline, steering and suspension components/fasteners should exhibit no damage or looseness.

Inflation pressure of both front and rear tires should be at the recommended setting of 241 kPa (35 psi) cold.

Tires should not exhibit uneven wear or excessive runout and wheels should be undamaged and true. Maximum runout at the tire should not exceed 1.57mm (0.062″) radial or 1.78mm (0.070 ″) lateral. Refer to Workshop Manual Section 100-04 for Diagnostic Procedure.

NOTE
TIRES THAT ARE UNEVENLY WORN OR HAVE EXCESSIVE RUNOUT SHOULD BE REPAIRED OR REPLACED USING NORMAL SERVICE PROCEDURES.

Tire/wheel assemblies should be balanced on a calibrated balancer only after driving the vehicle to remove any temporary “flat spotting” of the tires that may have occurred while the vehicle was parked. A drive of 8-16 km (5-10 miles) at highway speeds is recommended to get the tires to normal operating temperature.

NOTE
USE OF A BALANCER THAT CAN MEASURE ROAD FORCE VARIATION (ROTUNDA GSP9700 OR EQUIVALENT) IS STRONGLY RECOMMENDED TO OPTIMIZE MATCHING OF THE TIRES TO THE WHEELS.

Road test the vehicle (with the customer, if possible) to verify the condition (a slight, wheel-speed vibration above 72 km/h (45 mph) on smooth roads that can be felt in the steering and/or in the floor pan and front seat). If an unacceptable level of vibration is detectable, perform the following Service Procedure.

SERVICE PROCEDURE

Locate a tire balancer that measures force variation and measure force on each tire. If a force variation balancer is not available, rotate the tires and/or swap tires/wheels with a known good vehicle to isolate the tire(s) causing the vibration.

Replace the suspect tire(s) using the normal tire replacement process (ATW).

NOTE
REPLACE ONLY THE TIRE(S) THAT HAVE BEEN IDENTIFIED AS CAUSING THE VIBRATION. IT IS UNLIKELY THAT ALL 4 TIRES NEED TO BE REPLACED.

NOTE
IF THE TIRES ARE MOUNTED BY AN OUTSIDE SOURCE (e.g., TIRE DEALER), ENSURE THAT THE FOLLOWING TIRE-TO-WHEEL INDEXING PROCEDURE IS PROVIDED TO THE FACILITIES.

1. Mount the new tires using the following procedure:

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NOTE
THE TIRE SIDEWALL WITH THE YELLOW INDEX STICKER AND/OR THE CONICITY STAMP MARKINGS MUST BE INSTALLED IN THE OUTBOARD POSITION ON THE WHEEL.

a. Index the tires to the wheels. Indexing is necessary to obtain the best possible ride quality. Align the yellow index sticker on the tire (Figure 1) with the sprayed yellow paint mark located in the "dropwell" area of the wheel (Figure 2). If the yellow index sticker on the tire is not present, align the circle of the white conicity stamp marking on the outer portion of the sidewall (Figure 3) with the yellow paint mark in the dropwell area of the wheel.

b. Inflate the tire/wheel assemblies to 300 kPa (44 psi) to fully seat the tire beads on the wheel. Then reduce the pressure to the recommended setting of 241 kPa (35 psi) cold and balance the assemblies.

2. Install tire/wheel assemblies on vehicle and road test.

CLAIMING INSTRUCTIONS

- **Program Code:** None
- **Causal Part:** Tire
- **Condition Code:** 42
- **Part Number:** Use Tire Program Part #
- **Labor Operation:** Use Published Labor Ops. (Or Actual Time If None Are Available)
- **Warranty Status:** Handle As A Normal Service Repair, Replacing Only Those Tires Exhibiting Unacceptable Characteristics.

NOTE
DOT CODES MUST BE ENTERED IN THE DTC FIELDS FOR BOTH NEW AND REPLACED TIRES.

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-18-1
OASIS CODES: 303000, 306000, 703000, 703300

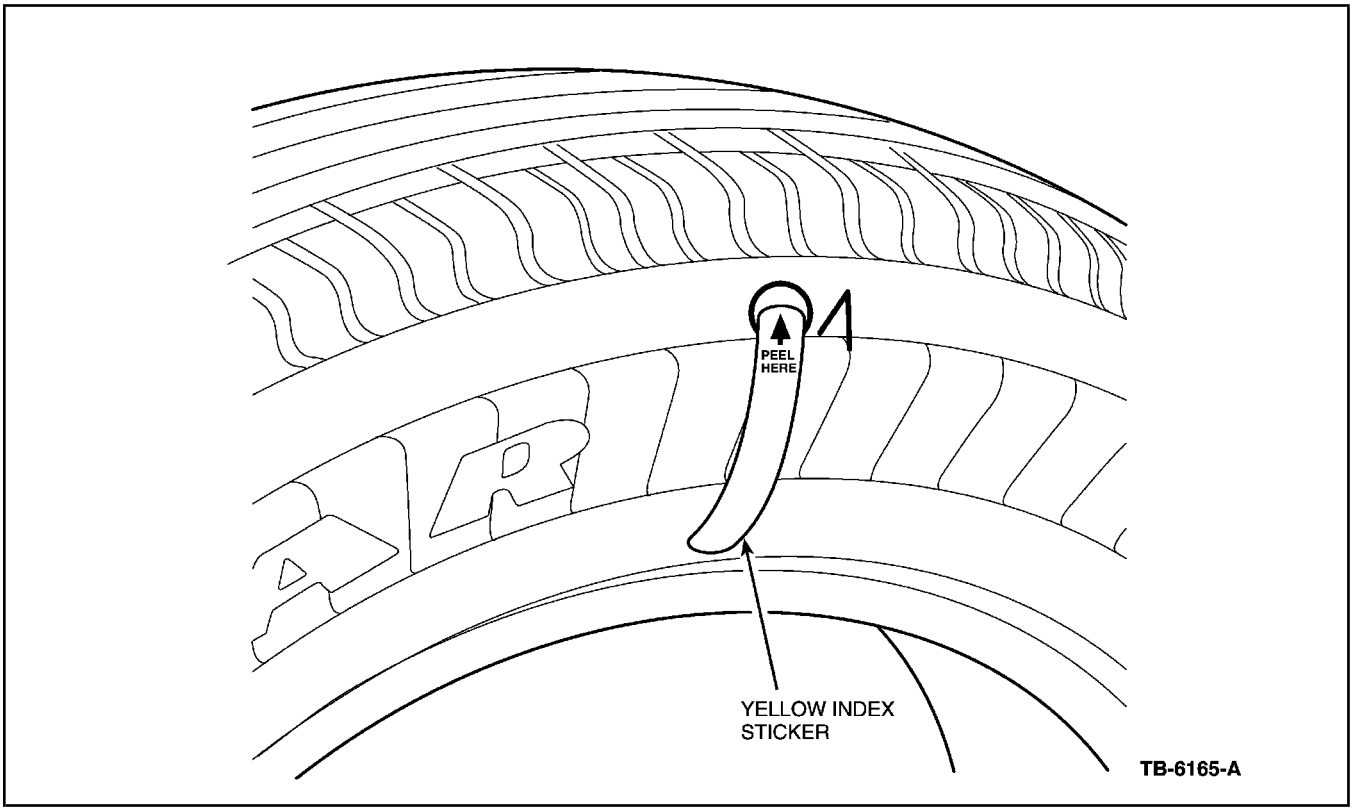


Figure 1 - Article 01-22-2

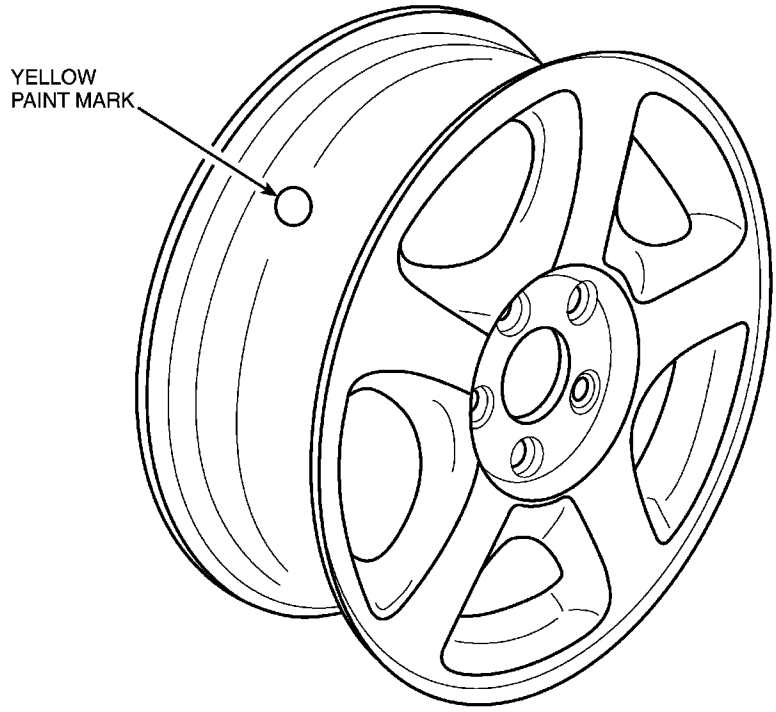


Figure 2 - Article 01-22-2

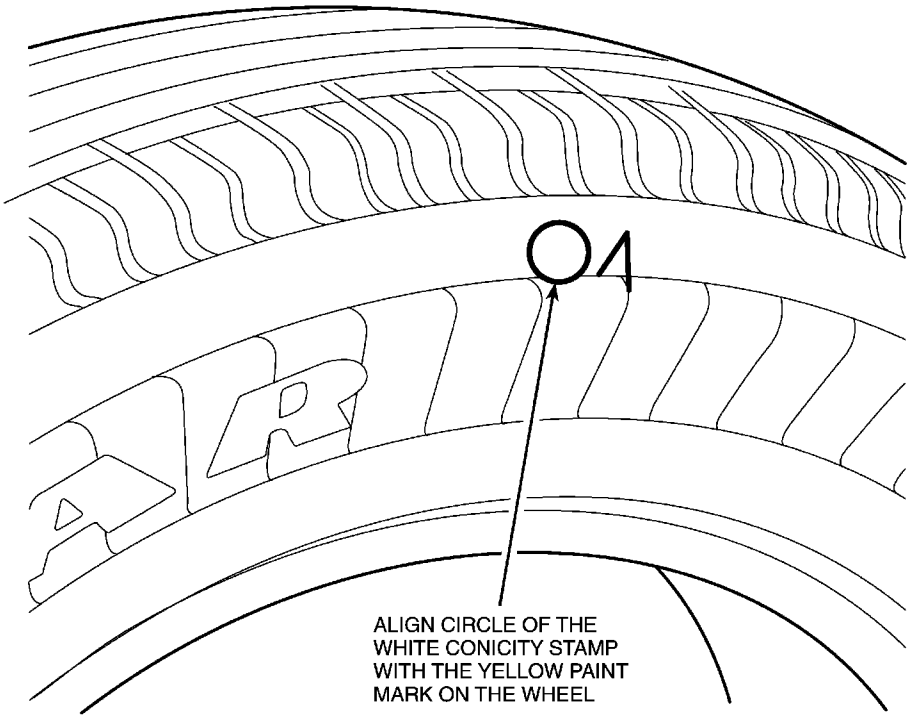


Figure 3 - Article 01-22-2